Educational Psychology Review, Vol. 6, No. 3, 1994

Why Innovate with Education?

Earl C. Butterfield[1,2]

Pressley and Harris have correctly highlighted many unfortunate limits of educational intervention research and institutions for disseminating research reports. They have suggested useful changes in how intervention research is done, and, except possibly for their emphasis on faithful replication, they have argued correctly that the changes are feasible. Unfortunately, they represent educational intervention research as being important primarily as guidance for educational reformers. I argue that research does not guide educational intervention, and that even if it is improved in the required ways they suggest it will not guide educational innovation. Instead, the valuable function of their analysis is that it could improve the scientific quality of educational research.

KEY WORDS: educational innovation; theory testing; research dissemination.

INTRODUCTION

I believe that when Pressley and Harris say "educational intervention research" they mean research that evaluates teaching that might increase what and how much at least some students learn in school. They mean research that, if done well, would provide a rational basis for educational innovation and reform.

I asked myself whether any research, regardless of its quality, can and should drive educational innovation? I answered that I do not believe any research has a credible chance of influencing educational innovation. Educational innovation fails for many complex reasons (Sarason, 1990), compared to which incompletely reported internally and externally invalid research pales in significance. Even if as Pressley and Harris advocate all

[1]University of Washington.
[2]Correspondence should be directed to Earl C. Butterfield, University of Washington.

1040-726X/94/0900-0209$07.00/0

intervention research was designed by collaborators with starkly different beliefs about what kind of teaching works, fully reported, incisively criticized by intelligent commentators, and repeatedly replicated, the quality of educational programs in most if not all schools would remain as they are. Educational reform is not inspired or guided by research. Researchers may wish it were. They may even believe it sometimes is. But I am a researcher who believes the educational innovation is not importantly influenced by research.

So why comment on a paper about how to improve educational intervention research? First, a sound knowledge base will be needed if educational reform ever does draw on research. Second, legislative and other policy enablers do sometimes seek information and welcome testimony about practices that could improve education. Third, despite the spin that Pressley and Harris put on their thoughtful analysis of intervention studies, namely the spin that practice will be improved by better applied research, the important function of educational intervention research is to validate theory about the kinds of thinking and knowledge that underlie learning and problem solving.

My chief criticism of the paper by Pressley and Harris is that it contributes to the neglect of the scientific and theoretical reason for instructional research by over-selling its utility for the applied purpose of educational reform. As counterpoint to their argument, I emphasize that the best of what they call educational intervention research rests on theory in the form of detailed analyses of thinking that can best be simulated by explaining and exemplifying how to do it (Butterfield, Siladi, and Belmont, 1980; Butterfield and Dickerson, 1976). The argument is that telling and showing children how to think influences their thinking more precisely than manipulating stimulus materials, requiring particular responses, or administering drugs. Moreover, computer simulations of cognitive theory are not fully adequate alternatives to instructional experiments. Although computer simulation can establish the sufficiency of a cognitive theory's precise details, it does not allow the claim of having manipulated a cognitive process in humans (cf, Holtzman *et al.,* 1976). It follows that instructional tests of theory are indispensable to a cognitive science that would be experimentally rigorous and psychologically valid. I believe this is the defensible reason to accept and act on Pressley and Harris's cogent recommendations for improving intervention research.

Pressley and Harris report the opinions of experts on educational intervention research. In effect, Pressley and Harris provide commentary of others in their own article on the state of educational intervention research. I agree wholeheartedly with almost of all their assessment of what needs improvement, and I have only the few following quibbles on recommendations they make to improve the enterprise.

CONSTRUCTING CREDIBLE COMPARISON CONDITIONS

To increase the credibility of conditions that researchers compare to their preferred interventions, Pressley and Harris recommend consulting or, better yet, collaborating with designers of the compared conditions. The purpose of this recommendation is to eliminate confoundings that make it impossible to know whether observed differences among compared conditions are "due to the treatments as defined by the developers or due to differences between conditions that are irrelevant to the conceptual differentiation of the treatment conditions." I applaud the purpose of focusing research as precisely as possible on critical variables that differentiate conditions, but I balk at the idea that the best way to do this is to accept the judgment of interventions' designers. I balk for the same reasons Pressley and Harris give for not relying on innovators as reviewers of manuscripts comparing their innovations to others' innovations: bias that sometimes clouds an advocate's view of the intervention that he should know best. Furthermore, given the lamentable state of much intervention research, it probably makes less difference that compared conditions be faithful to the views and intents of their designers than that they be faithful to the tenor of the arguments advanced by Pressley and Harris. Only when the process of consulting and collaborating is sharply focused on obtaining the most interpretable results, and I am not idealistic enough to believe that would happen often, is it a better process than the concerted efforts of a like-minded team doing its best to realize the goals of Pressley and Harris's recommendations.

TIMELY REPORTS OF ALL INTERVENTION DETAILS

Pressley and Harris correctly note that convention presentations are too short to allow full description of details of the conditions and data analyses of educational intervention. To reduce the likelihood that such reports will fuel ineffective interventions, Pressley and Harris call on presenters to quickly publish full details of their interventions and their analysis. Given the realities of life as a researcher, such pleas will probably not solve the problem. Perhaps it would solve more of the problem if would-be interveners and convention organizers are also called upon to do their part.

Interveners could treat convention presentations as billboards, as an opportunity to learn who to contact for more information before intervening. Convention program committees could accept only intervention talks that are accompanied by detailed written materials insuring than any listener can secure needed details. Along with Pressley and Harris's suggestion, these two could guarantee that convention billboards advertise available products that interested consumers can obtain when they try.

TRUTH IN ADVERTISING INTERVENTIONS

Pressley and Harris call for scientists to restrict their claims about educational interventions to "ones that strictly follow from their designs and outcome." The point is to reduce the possibility that practitioners will implement programs that cannot attain their goals. In addition to the fact that this call arrogates more influence to scientists than they have on the implementation of educational interventions, it ignores the fact that their can be positive benefits to describing one's findings at a higher level of generality than "ones that strictly follow . . .." Often scientists want to stimulate research to follow up their findings, which is a legitimate purpose. It is easy to believe that others are more inspired to follow-up work that is described at a level of generality that makes it easy for them to relate it to their situation. How often are readers inspired by a discussion that sticks strictly to thoroughly validated arguments? If Pressley and Harris had spelled out what they mean by claims that strictly follow, I believe that hardly anyone would agree with their call. The number of measurement caveats alone that would need to be added to claims about any intervention research would daunt even sophisticated scientists and thoroughly confuse many would-be researchers. Nobody would be inspired to do any follow-up that conforms to the demanding but necessary criteria set forth so cogently by Pressley and Harris.

REPLICATIONS SHOULD BE PUBLISHED

Pressley and Harris are correct that successful and unsuccessful replications of important intervention studies should be published, but they seldom are. It is probably pie-in-the-sky to hope that they will be unless there are major changes in publication policy. A simple to implement policy would be for editors to publish only intervention studies that include an independent replication, including replications that yield different conclusions. The cost of such a simple policy could be staggering. Fewer intervention studies would be done. The cost of those that are done would be multiplied, probably by more than two if one means by independent replications that are done in different settings by different investigative teams. Statisticians would howl "Not necessary!" Inferential statistics are largely ways of estimating the probability of successful replication. Another simple to implement policy would be for granting agencies to fund intervention research only if applicants promise independent replications. The number of grants awarded would surely decrease by some large fraction, and many would argue that the decrease was not justified by the gains from replication. Imagine the quagmire of arguments that result from these two policy changes!

CONCLUSIONS

Pressley and Harris have correctly highlighted many unfortunate limits of educational intervention research and institutions for disseminating research reports. They have suggested useful changes in how intervention research is done, and, except possibly for their emphasis on faithful replication, they have argued correctly that the changes are feasible.

I am less sanguine about their suggestions for disseminating research. Their suggestions about dissemination assume that researchers, convention programmers, and editors believe that an important purpose of intervention research is to improve educational practice. I do not believe this, and I have been an editor, convention planner, and researcher. Rather, I believe that educational intervention is an extremely complex, political, and economic game played often by people whose sincere desire to have research improve education must compete with many other goals: Securing recognition as a scientist, building a convention program that will succeed in attracting enough registrants to pay its cost, and filling journal pages with timely papers, to name only a few.

I do believe that intervention research has extremely important scientific value that more than justifies Pressley and Harris's suggested improvements in its conduct. I am greatly pleased to have been asked to comment on their often wise remarks.

REFERENCES

Butterfield, E. C., and Dickerson, D. J. (1976). Cognitive theory and mental development. In Ellis, N. R. (ed.), *International Review of Research in Mental Retardation*. Academic Press, New York, pp. 105-137.

Butterfield, E. C., Siladi, D., and Belmont, J. M. (1980). Validating theories of intelligence. In Reese, H. and Lipsitt, L. P. (eds.), *Advances in Child Development and Child Behavior*. Academic Press, New York, pp. 96-162.
Holtzman, T. G., Pellegrino, J. W., and Glaser, R. (1976). Process training derived from a computer simulation theory. *Memory Cognit.* 4: 349-356.
Sarason, S. (1990). *The Predictable Failure of Educational Reform*. Jossey Bass, San Francisco, pp. xi-98.